HAC: Simple Hybrids, Smooth Schedules — Rethinking Adaptive Fusion for Resilient UAV Swarm Connectivity Recovery

Abstract

Self-healing connectivity recovery for unmanned aerial vehicle (UAV) swarms requires restoring network connectivity rapidly after partial node failures while preserving wide communication coverage. Peng et al. (2026) recently addressed this trade-off with R2C, combining an intra-subnetwork virtual-force expansion module, a subnetwork-level multipartite graph convolution module, and a time-adaptive weighted fusion schedule (AWF), yet bundling these components obscures the individual roles of spatial reasoning granularity and schedule transition steepness. We systematically ablate this design space across decision paradigm, spatial reasoning granularity, and connectivity–coverage trade-off strategy through five-seed controlled experiments. We propose Hybrid Adaptive Convolution (HAC), which fuses a parameter-free virtual-force heuristic branch with a single global graph convolution branch at the velocity level while omitting subnetwork partitioning, convex-hull boundary extraction, and multipartite graph construction. Across random and uniform deployments, HAC boosts the 80%-failure recovery rate by up to 76.4 percentage points (pp) over single-branch learned baselines and retains 88.4% to 91.2% of R2C's reported spatial coverage (within 5.9 to 7.9 pp). Furthermore, we show that transplanting R2C's steep adaptive schedule directly into HAC degrades high-damage recovery, whereas smoothing the transition slope restores robustness and improves coverage uniformity. By decoupling training-time and evaluation-time schedules, we prove that preserving a test-time heuristic safety margin prevents late-stage link disruption while smooth training schedules act as an implicit residual curriculum that halves cross-seed variance. Finally, a subnetwork-level hybrid counter-experiment (HSC) demonstrates that schedule smoothing rescues moderate-damage recovery but cannot overcome rigid-translation expressiveness limits under severe fragmentation.

1 Introduction

1.1 Problem and Motivation

Partial node failures fracture swarm topologies and threaten mission continuity. Unmanned aerial vehicle (UAV) swarms provide rapid, flexible wireless coverage in post-disaster relief, dynamic battlefields, and congested areas (Coletta et al., 2023; Shi et al., 2025; Wang et al., 2022, 2024b). During operation, internal faults or external attacks can disable a subset of UAVs at any time (Wang et al., 2024a; Kurunathan et al., 2023). When substantial node losses occur, the communication graph fractures into disconnected subnetworks, sharply degrading both communication reliability and spatial coverage across the mission area (Li et al., 2023; Xu et al., 2022; Peng et al., 2022). The self-healing communication connectivity (SCC) problem seeks a distributed or partially observed velocity control policy that repositions surviving UAVs to restore full graph connectivity as rapidly as possible while maximizing spatial coverage.

Connectivity restoration and spatial coverage preservation are in direct physical tension. Pulling surviving UAVs toward the swarm center accelerates reconnection among fragmented components but shrinks the covered footprint, whereas spreading nodes outward preserves area coverage at the risk of slowing or preventing reconnection. Balancing these competing forces becomes especially challenging when large failure ratios leave small, widely separated subnetworks across the mission space.

1.2 Limitations of Existing Approaches

Hand-crafted heuristic rules fail to scale under severe, unstructured failures. Non-learning SCC algorithms either identify and replace critical cut-vertices through cascaded neighbor movements (Akkaya et al., 2009; Mi and Yang, 2015; Zhang et al., 2024) or guide surviving nodes toward predefined rendezvous points using potential fields (Joshi, 2012; Shriwastav, 2018; Chen et al., 2020; Wang et al., 2024c). Although heuristic rules require no training data and preserve local spacing well, their static coordination assumptions break down when severe node losses disconnect components beyond local sensing ranges (Tosun et al., 2024).

Purely learned graph policies suffer from spatial collapse or rigid-body expressiveness limits. Graph neural network (GNN) methods formulate SCC as a learned graph optimization task and reconnect fragmented topologies faster than heuristics (Mou et al., 2022; Peng et al., 2025; Wang et al., 2025b). However, message-passing aggregation biases surviving UAVs to cluster around high-degree nodes, causing severe spatial collapse and coverage loss. Diffusion-based and differential-subgraph convolutions (Lin et al., 2024; Lin and Ding, 2025) encourage broader dispersion, yet struggle to maintain stable layouts without explicit local repulsion.

1.3 Our Approach and Findings

Isolating spatial granularity and schedule steepness reveals what makes hybrid recovery work. Peng et al. (2026) recently introduced R2C, which combines a buffered virtual-force field for intra-subnetwork expansion (ISE) with a multipartite graph convolution module for inter-subnetwork translation (IST) under an adaptive weighted fusion (AWF) schedule. While R2C achieves strong empirical performance, its module-level ablations toggle entire branches on and off without isolating whether the learned component must operate at subnetwork granularity or whether AWF requires its specific steep sigmoid transition. To answer both questions, we introduce **HAC** (Hybrid Adaptive Convolution), a streamlined architecture that fuses R2C's parameter-free virtual-force branch at the velocity level with a single global k-nearest-neighbor (k-NN) graph convolution branch, eliminating subnetwork partitioning, convex-hull boundary extraction, and multipartite virtual edges.

Five-seed experiments across random and uniform deployments show that fixed-weight global hybridization (**HAC-Fixed**) and smooth adaptive hybridization (**HAC-Smooth**) achieve near-perfect connectivity recovery through 60% failure and retain 88.4% to 91.2% of R2C's reported coverage at 80% failure. Conversely, directly applying R2C's steep adaptive schedule ($k = 30, s_0 = 0.125$) to HAC degrades 80%-failure recovery from 46.6/50 to 14.2/50. Our five-seed schedule sweep and train–evaluation decoupling study reveal two complementary mechanisms: (i) at deployment time, a steep sigmoid zeroes out the heuristic branch too early, stripping away its local attractive-zone descent guarantee and repulsive spacing buffer; and (ii) during differentiable rollout training, a smooth transition acts as an implicit residual curriculum that reduces cross-seed variance by more than half when paired with balanced evaluation weights (47.4 ± 1.9 out of 50). Substituting a subnetwork-level rigid-translation branch (**HSC**) confirms that schedule smoothing rescues moderate fragmentation

(60% failure) but cannot overcome the lack of node-level velocity expressiveness at 80% failure.

1.4 Contributions

1. **Three-axis design taxonomy (§3, §4).** We categorize SCC methods along decision paradigm, spatial reasoning granularity, and connectivity–coverage trade-off strategy, and demonstrate across five random seeds that purely global learned policies (GAC) suffer from spatial clustering whereas purely subnetwork-level rigid policies (SAC) fail deterministically at 60%–80% node loss.
2. **Minimalist global hybrid architecture (§5, §8).** We propose HAC, fusing a parameter-free virtual-force branch with a global k-NN graph convolution branch. Five-seed evaluations across two deployment geometries show that HAC-Fixed and HAC-Smooth outperform single-branch learned baselines by up to 76.4 pp in 80%-failure recovery rate while retaining 88.4% to 91.2% of R2C's published coverage without boundary-hull or multipartite machinery.
3. **Theoretical and empirical decomposition of adaptive fusion schedules (§6, §9).** We prove a local attractive-zone descent guarantee for the heuristic branch, formulate a high-autonomy exposure framework, and decouple training vs. evaluation schedules across five seeds to quantify test-time safety-margin protection alongside training-time curriculum stabilization, validated against a five-seed subnetwork-level hybrid counter-experiment (HSC).
4. **Count-invariant connectivity loss and reproducible benchmark (§5, Appendix A).** We identify and correct a pair-count bias in log-sum-exp soft-minimum connectivity surrogates, enforce full-stack random seeding, and release the complete multi-seed simulation and training suite.

2 Related Work

2.1 Non-Learning Self-Healing Methods

Critical-node compensation scales poorly with failure density. Early topological recovery schemes detect cut-vertices whose removal partitions the network and trigger cascaded neighbor movements to fill the resulting coverage gaps (Abbasi and Younis, 2008; Akkaya et al., 2009; Mi and Yang, 2015; Zhang et al., 2024). While effective for isolated single-node faults, identifying critical nodes and coordinating multi-hop cascades incur high signaling overhead that fails when simultaneous failures eliminate multiple adjacent relays.

Rendezvous-point aggregation struggles under uncertain fragmentation. Alternative heuristic methods guide surviving UAVs or disconnected clusters toward predetermined rendezvous coordinates using Coulomb-force models (Joshi, 2012), round-table negotiation (Shriwastav, 2018), or artificial potential fields (Chen et al., 2020; Wang et al., 2024c). Although parameter-free and computationally lightweight, fixed attraction rules cannot adapt when surviving subnetworks are separated beyond their local communication horizon (Tosun et al., 2024).

2.2 Learning-Based Self-Healing Methods

Relaxed GNN and attention policies accelerate reconnection but induce spatial collapse. Casting SCC as a differentiable graph optimization problem enables data-driven policies to reason over long-range topological structure (Karalias and Loukas, 2020; Peng et al., 2025). Mou et al. (2022) apply a relaxed graph convolutional network over virtual links, and Guo et al. (2024) incorporate spatiotemporal multi-head attention. Despite faster reconnection than heuristics, message

aggregation pulls surviving UAVs tightly toward high-degree hubs, causing severe spatial clustering and coverage loss.

Diffusion and differential-subgraph convolutions lack explicit local collision buffering. Recent learning architectures introduce multi-hop diffusion (Lin et al., 2024) or bipartite convolution over differential subgraphs (Lin and Ding, 2025) to promote dispersed recovery. While these designs improve post-recovery spread compared to standard GCNs, they do not enforce hard inter-agent spacing zones during transit.

2.3 Hybrid Heuristic–Learning Architectures and Positioning

Prior hybrid frameworks either fix node-level weights or bundle subnetwork modules with steep schedules. Li et al. (2025) combine distributed multi-agent reinforcement learning with an artificial potential field at node-level granularity under fixed weighting, whereas Wang et al. (2025b) rely solely on graph reinforcement learning. Related multi-robot coverage and spectral GNN studies explore substructure-aware message passing (Kipf and Welling, 2016; Zeng et al., 2023; Liu and Wang, 2025; Mutzari et al., 2025; Ye et al., 2022). Peng et al. (2026) advance this line with R2C, pairing buffered virtual forces for intra-subnetwork expansion (ISE) with convex-hull boundary extraction (Barber et al., 1996) and Chebyshev multipartite graph convolution for inter-subnetwork translation (IST), blended via a sigmoid adaptive weighted fusion (AWF) schedule (Elfwing et al., 2018). Because R2C's ablations only remove entire modules, the literature lacks a controlled isolation of spatial reasoning granularity (global vs. subnetwork learned branches) and schedule transition steepness. Our work directly bridges this gap.

3 Preliminaries and Problem Formulation

3.1 UAV Swarm Network Model

Swarm connectivity is governed by the algebraic multiplicity of the communication Laplacian. Following Peng et al. (2026), at discrete time step t the surviving UAV swarm forms a temporal proximity graph $G_t = (V_t, E_t, P_t)$, where $V_t = \{v_{1,t}, \ldots, v_{|V_t|,t}\}$ denotes the surviving UAVs, $P_t = \{\mathbf{p}_{1,t}, \ldots, \mathbf{p}_{|V_t|,t}\}$ with $\mathbf{p}_{i,t} \in \mathcal{A} \subset \mathbb{R}^2$ denotes their planar coordinates within mission area $\mathcal{A}$, and E_t denotes active communication links. A direct air-to-air link (Ma et al., 2019; Wang et al., 2025a) exists between v_i and v_j if and only if their Euclidean distance $d_{ij,t} = \|\mathbf{p}_{i,t} - \mathbf{p}_{j,t}\|_2$ satisfies $d_{ij,t} \leq R_c$, where the communication radius R_c follows a free-space path-loss model (Ni et al., 2024). Let $\mathbf{A}_t$ be the adjacency matrix and $\mathbf{L}_t = \mathbf{D}_t - \mathbf{A}_t$ be the graph Laplacian. The number of connected components C_t equals the multiplicity of the zero eigenvalue of $\mathbf{L}_t$, so the swarm is fully connected if and only if $C_t = 1$. Each UAV also covers a disk $\mathcal{C}_i(t)$ of radius R_{cov}, yielding the collective area coverage ratio $C_{\text{cov}}(t) = |\bigcup_{i \in V_t} \mathcal{C}_i(t)|/|\mathcal{A}|$.

3.2 Problem Definition

The SCC objective jointly penalizes normalized reconnection time and post-recovery coverage loss. Let t_1 denote the instant of node failure and $t_r \leq t_1 + T$ denote the earliest subsequent step at which $C_{t_r} = 1$ within horizon T. The normalized recovery time ratio is $T_{\text{rec}} = (t_r - t_1)/T$. The control policy generates bounded per-node velocities $\hat{V}_t = \{\hat{\mathbf{v}}_{1,t}, \ldots, \hat{\mathbf{v}}_{|V_t|,t}\}$ to minimize

$$J_c = T_{\text{rec}} + \big(1 - C_{\text{cov}}(t_r)\big), \tag{1}$$

Table 1: Taxonomy of the $3 \times 3 \times 3$ SCC design space positioning prior methods and our four controlled architectures (GAC, SAC, HAC, and HSC).

Paradigm	Granularity	Trade-off	Representative method(s)
Heuristic	Node-level	Fixed	Akkaya et al. (2009); Mi and Yang (2015); Wang et al. (2024c)
Heuristic	Subnetwork	Fixed	Chen et al. (2020)
Heuristic	Global	Fixed	Lin and Ding (2025)
Heuristic	Global	Staged	HERO (Mi et al., 2011)
Learned	Node-level	Fixed	Mou et al. (2022)
Learned	Global	Fixed	Guo et al. (2024); **GAC (this work, §4)**
Learned	Subnetwork	Fixed	Lin et al. (2024); **SAC (this work, §4)**
Hybrid	Node-level	Fixed	Li et al. (2025)
Hybrid	Node+Subnet	Adaptive	R2C (ISE+IST+AWF) (Peng et al., 2026)
Hybrid	**Global**	**Fixed / Adaptive**	**HAC-Fixed / HAC-Smooth (this work, §5)**
Hybrid	**Subnetwork**	**Fixed / Adaptive**	**HSC (this work, §9)**

subject to four operational constraints: (C1) terminal connectivity $C_{t_r} = 1$ within horizon T, where trials failing to reconnect by $t_1 + T$ are marked unsuccessful and evaluated primarily via recovery success rate; (C2) speed and boundary limits $\|\hat{\mathbf{v}}_{i,t}\|_2 \leq \hat{v}_{\max}$ and $\mathbf{p}_{i,t} \in \mathcal{A}$; (C3) discrete kinematics $\mathbf{p}_{i,t+1} = \mathbf{p}_{i,t} + \Delta t \cdot \hat{\mathbf{v}}_{i,t}$; and (C4) link feasibility $d_{ij,t} \leq R_c$. Minimizing T_{rec} pulls components inward whereas minimizing $1 - C_{\text{cov}}(t_r)$ pushes nodes outward.

3.3 A Taxonomy of the Design Space

Three orthogonal design axes organize prior SCC methods and our controlled ablations. Every self-healing controller must specify three architectural choices:

- **Axis 1 (Decision Paradigm):** Whether motion commands originate from parameter-free physical or graph rules (*heuristic*), a trained parametric network (*learned*), or a velocity-level blend of both (*hybrid*).
- **Axis 2 (Spatial Reasoning Granularity):** Whether the controller reasons over 1-hop communication neighbors (*node-level*), treats each connected component as a rigid unit (*subnetwork-level*), or passes messages across the entire surviving swarm (*global-level*).
- **Axis 3 (Trade-off Strategy):** Whether connectivity and coverage priorities are combined with constant weights (*fixed*), executed in strict temporal phases (*staged*), or continuously re-weighted over time (*adaptive*).

Table 1 maps representative literature and the four experimental architectures studied in this work into this $3 \times 3 \times 3$ taxonomy: **GAC** (global learned policy), **SAC** (subnetwork-level rigid-translation learned policy), **HAC** (our proposed hybrid combining node-level heuristic forces with a global learned GCN under fixed or smooth adaptive schedules), and **HSC** (a hybrid counter-experiment pairing the same heuristic branch with SAC's subnetwork-level learned branch).

Table 2: Motivating single-branch learned baselines under random deployment (mean ± std over 5 seeds, 50 trials per seed). GAC-Fixed exhibits spatial clustering (elevated AND), whereas SAC-Fixed avoids clustering but fails deterministically at 60%–80% node loss.

Failure	GAC-Fixed (global learned)		SAC-Fixed (subnetwork learned)	
rate	AND	Success/50	AND	Success/50
20%	13.68 ± 0.28	49.0 ± 0.6	6.71 ± 0.09	29.4 ± 3.4
40%	12.48 ± 0.61	47.2 ± 1.7	5.32 ± 0.18	8.8 ± 1.9
60%	10.55 ± 0.78	42.8 ± 3.7	n/a	0.0 ± 0.0
80%	6.00 ± 0.26	19.8 ± 5.5	n/a	0.0 ± 0.0

4 Motivating Experiments: Failure Modes of Single-Branch Designs

Single-branch learned policies face a fundamental dilemma between spatial collapse and fragmentation failure. Before introducing our hybrid architecture, we evaluate two pure learned baselines that isolate the spatial granularity axis under identical loss functions and training rollouts:

- **Global Adaptive-weighted Convolution (GAC-Fixed):** A 4-layer GCN operating over a Euclidean k-NN graph ($k = 8$) across all surviving UAVs regardless of communication range R_c, predicting per-node 2D velocities directly.
- **Subnetwork-level Adaptive Convolution (SAC-Fixed):** A 4-layer GCN operating over a k-NN graph of connected-component centroids (pooled from component coordinates and relative sizes), predicting a single rigid 2D translation velocity shared by all UAVs within each subnetwork.

Global message passing causes severe spatial clustering, whereas rigid subnetwork translation collapses under moderate-to-high node loss. Table 2 summarizes five-seed evaluations under random deployment. GAC-Fixed maintains moderate connectivity recovery up to 60% node failure but produces an elevated trajectory-averaged node degree (AND of 13.68 ± 0.28 at 20% failure, more than double SAC-Fixed's 6.71 ± 0.09), confirming that unconstrained global GNN aggregation pulls UAVs into dense clusters at the expense of area coverage. Conversely, forcing all nodes within a component to translate rigidly in SAC-Fixed prevents internal clustering and keeps AND low (6.71 at 20% and 5.32 at 40%), but destroys recovery capability as fragmentation increases, dropping to 8.8 ± 1.9 recoveries out of 50 at 40% failure and 0.0 ± 0.0 across all five seeds at 60% and 80% failure. Because isolated single-UAV and small-cluster subnetworks cannot reshape their internal boundaries to bridge wide gaps under rigid translation, neither granularity succeeds alone. This complementary failure pattern directly motivates fusing a local spacing heuristic with a global learned velocity field.

5 Method: Hybrid Adaptive Convolution (HAC)

5.1 Overview and Architectural Minimalism

HAC pairs parameter-free local virtual forces with a single global graph convolution at the velocity level. To test whether R2C's complex subnetwork machinery is necessary, HAC instantiates the *hybrid* × *global* cell of Table 1 in its simplest form. Each surviving UAV's velocity command is synthesized from two parallel branches: a deterministic virtual-force heuristic branch

that regulates 1-hop neighbor spacing and boundary avoidance, and a global k-NN GCN branch that provides long-range directional guidance across disconnected components without convex-hull boundary extraction or multipartite virtual edges.

5.2 Heuristic Branch: Buffered Virtual-Force Field

A three-zone virtual spring regulates local spacing and prevents boundary escape without learnable parameters. Adopting R2C's buffered Hookean force model (Rychlewski, 1984; Peng et al., 2026) with neutral zone $[d_l, d_u] = [0.8R_c, 0.9R_c] \subset (0, R_c]$, the pairwise force exerted on UAV v_i by a communication neighbor $v_j \in \mathcal{N}_i = \{j : 0 < d_{ij,t} \le R_c\}$ is

$$\mathbf{F}'_{ij,t} = \begin{cases} -\exp\left(-\alpha \cdot \dfrac{d_{ij,t}}{d_l}\right) \mathbf{u}_{ij,t}, & 0 < d_{ij,t} < d_l, \\ \mathbf{0}, & d_l \le d_{ij,t} \le d_u, \\ \exp\left(-\beta \cdot \dfrac{R_c - d_{ij,t}}{R_c - d_u}\right) \mathbf{u}_{ij,t}, & d_u < d_{ij,t} \le R_c, \end{cases} \tag{2}$$

where $\mathbf{u}_{ij,t} = (\mathbf{p}_{j,t} - \mathbf{p}_{i,t})/d_{ij,t}$ is the unit vector pointing from v_i to v_j, and $\alpha = \beta = 0.01$. To prevent UAVs from drifting outside the mission square $\mathcal{A} = [0, L]^2$, any UAV within buffer distance $d_e = 0.15R_c$ of a boundary edge $b \in \mathcal{B} = \{\text{left, right, bottom, top}\}$ receives an inward repulsion force $\mathbf{F}''_{i,t} = \sum_{b \in \mathcal{B},\, 0 \le d_{i,b} < d_e} \exp(-\delta \cdot d_{i,b})\, \mathbf{n}_b$, where $d_{i,b} \ge 0$ is the perpendicular distance to edge b, $\mathbf{n}_b$ is its inward unit normal, and $\delta = 0.01$. Summing over communication neighbors yields $\mathbf{F}_{i,t} = \sum_{j \in \mathcal{N}_i} \mathbf{F}'_{ij,t} + \mathbf{F}''_{i,t}$ and normalized heuristic velocity $\mathbf{v}'_{i,t} = \hat{v}_{\max} \cdot \mathbf{F}_{i,t}/(\|\mathbf{F}_{i,t}\|_2 + \epsilon)$.

5.3 Learned Branch: Global Graph Convolution

A global Euclidean k-NN graph enables cross-component message passing without subnetwork partitioning. At each step t, we construct a symmetric k-NN graph ($k = 8$) over the Euclidean coordinates of all surviving UAVs so that topological information propagates across disconnected components even when inter-component distances exceed R_c. Each UAV v_i is represented by a 3-dimensional feature vector $\mathbf{x}_{i,t} = [p_{i,x}/L,\ p_{i,y}/L,\ \deg_k(v_i)/\max_j \deg_k(v_j)]^\top$, omitting explicit subnetwork IDs so that node representations remain invariant to arbitrary component labelings. Passing $\{\mathbf{x}_{i,t}\}$ through an $L_{\text{gcn}} = 4$ layer GCN (Kipf and Welling, 2016) with hidden dimension 128 and ReLU activations, followed by a two-layer MLP and elementwise tanh scaling by $\hat{v}_{\max}$, produces the learned velocity vector $\mathbf{v}''_{i,t}$.

5.4 Velocity-Level Fusion: Fixed and Smooth Adaptive Schedules

Velocity-level normalization blends local heuristic expansion with global learned translation. Given time horizon H ($H = T_{\text{train}}$ during training and $H = T$ during evaluation), the adaptive fusion schedule assigns weights $w_T(t)$ and $w_E(t)$ via a parameterized sigmoid:

$$w_T(t; k, s_0) = \frac{1}{1 + \exp\big(-k \cdot (t/H - s_0)\big)}, \qquad w_E(t; k, s_0) = 1 - w_T(t; k, s_0), \tag{3}$$

where $k > 0$ governs transition steepness and $s_0 \in (0, 1)$ sets the midpoint fraction. The two branch velocities are fused and renormalized to maximum speed $\hat{v}_{\max}$:

$$\hat{\mathbf{v}}_{i,t} = \hat{v}_{\max} \cdot \frac{w_E(t)\, \mathbf{v}'_{i,t} + w_T(t)\, \mathbf{v}''_{i,t}}{\left\| w_E(t)\, \mathbf{v}'_{i,t} + w_T(t)\, \mathbf{v}''_{i,t} \right\|_2 + \epsilon}. \tag{4}$$

Algorithm 1 HAC End-to-End Differentiable Rollout Training (one epoch)

Require: Learned policy π_θ, rollout horizon T_{train}, fusion schedule $w_T(\cdot)$

Sample failure ratio $r \in \{0.2, 0.4, 0.6, 0.8\}$; initialize surviving positions $\mathbf{P}_0$
$\mathcal{L}_{\text{total}} \leftarrow 0$
for $t = 0, \ldots, T_{\text{train}} - 1$ **do**
    Partition $\mathbf{P}_t$ into connected components $\{S_m\}$ under communication range R_c
    Compute parameter-free heuristic velocities $\mathbf{v}'_t$ via Eq. (2) ▷ stop gradient
    Construct global Euclidean k-NN graph on $\mathbf{P}_t$ and predict $\mathbf{v}''_t \leftarrow \hat{v}_{\max}\, \pi_\theta(\mathbf{P}_t)$
    Compute fused velocities $\hat{\mathbf{v}}_t$ from $(\mathbf{v}'_t, \mathbf{v}''_t, 1 - w_T(t), w_T(t))$ via Eq. (4)
    $\mathbf{P}_{t+1} \leftarrow \text{clip}(\mathbf{P}_t + \Delta t \cdot \hat{\mathbf{v}}_t,\ 0,\ L)$
    $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{total}} + 0.5\, \mathcal{L}_{\text{conn}}(\mathbf{P}_{t+1}) + 0.5\, \mathcal{L}_{\text{cov}}(\mathbf{P}_{t+1}) + \lambda\, \mathcal{L}_{\text{move}}(\mathbf{P}_t, \mathbf{P}_{t+1})$
end for
Backpropagate $\mathcal{L}_{\text{total}}/T_{\text{train}}$ and update θ via Adam with gradient clipping at 1.0

We instantiate two primary variants of our method: **HAC-Fixed**, which sets constant equal weights $w_E(t) = w_T(t) = 0.5$ across all steps, and **HAC-Smooth**, which employs a gentle handover ($k = 5, s_0 = 0.7$) in contrast to R2C's steep default ($k = 30, s_0 = 0.125$). We also evaluate a decoupled variant (**HAC-Curriculum**) that trains under the smooth schedule ($k = 5, s_0 = 0.7$) and deploys with fixed equal weights $w_E = w_T = 0.5$.

5.5 Count-Invariant Differentiable Training Objective

Subtracting log pair counts makes soft-minimum inter-component distance estimation invariant to cluster size. Because the discrete component count C_t is non-differentiable, we train the learned branch end-to-end through T_{train}-step differentiable rollouts using a corrected soft-minimum connectivity surrogate. For every pair of disconnected components (S_i, S_j) at step t with node counts $n_i = |S_i|$ and $n_j = |S_j|$, we define

$$\mathcal{L}_{\text{conn}}^{(i,j)} = -\tau_{\text{c}} \left(\log \sum_{u \in S_i,\, v \in S_j} \exp\left(-\frac{d_{uv,t}}{\tau_{\text{c}}} \right) - \log(n_i n_j) \right), \tag{5}$$

with temperature $\tau_{\text{c}} = 50\,\text{m}$, averaged over all $\binom{C_t}{2}$ component pairs. Without the $-\log(n_i n_j)$ correction, standard log-sum-exp systematically under-penalizes large components by $\tau_{\text{c}} \log(n_i n_j)$ and produces spuriously negative losses (verified numerically in Appendix A). Area coverage is approximated over Monte Carlo grid samples G_{mc} via $\mathcal{L}_{\text{cov}} = \frac{1}{|G_{\text{mc}}|} \sum_{\mathbf{g} \in G_{\text{mc}}} \big(1 - \sigma\big((R_{\text{cov}} - \min_i \|\mathbf{g} - \mathbf{p}_{i,t}\|_2)/\tau_{\text{v}}\big)\big)$ with $\tau_{\text{v}} = 10\,\text{m}$, and movement regularization is $\mathcal{L}_{\text{move}} = \frac{1}{|V_t|} \sum_i \|\mathbf{p}_{i,t+1} - \mathbf{p}_{i,t}\|_2$. The per-step loss is $\mathcal{L}_t = 0.5\, \mathcal{L}_{\text{conn}} + 0.5\, \mathcal{L}_{\text{cov}} + \lambda\, \mathcal{L}_{\text{move}}$ with $\lambda = 0.01$. Algorithms 1 and 2 detail the training and inference procedures.

6 Theoretical Analysis

6.1 Local Monotonicity of the Heuristic Branch

The heuristic virtual-force branch provides a certifiable local descent direction on inter-node distance within the attractive zone. We first analyze the test-time stability provided by the parameter-free branch.

Algorithm 2 HAC Deployment Rollout

Require: Trained policy π_θ, evaluation horizon T, evaluation schedule $w_T^{\text{eval}}(\cdot)$

for $t = 0, \dots, T-1$ **do**
  if $C_t = 1$ **then return** success at recovery step $t_r = t$
  end if
  Compute $\mathbf{v}'_t$ via Eq. (2) and $\mathbf{v}''_t \leftarrow \hat{v}_{\max}\, \pi_\theta(\mathbf{P}_t)$
  Update $\mathbf{P}_{t+1} \leftarrow \text{clip}\big(\mathbf{P}_t + \Delta t \cdot \text{fuse}(\mathbf{v}'_t, \mathbf{v}''_t, 1 - w_T^{\text{eval}}(t), w_T^{\text{eval}}(t)),\ 0,\ L\big)$
end for
return failure (horizon T exhausted with $C_T > 1$)

Proposition 1 (Attractive-zone descent guarantee)**.** *Let v_i, v_j be two surviving UAVs at distance $d_u < d_{ij,t} \le R_c$ in the attractive zone of Eq. (2). If no repulsive neighbor forces or boundary forces act on v_i and v_j, then applying the heuristic velocity alone ($w_E = 1, w_T = 0$) with step size $0 < \Delta t < d_{ij,t}/(2\hat{v}_{\max})$ strictly decreases their Euclidean separation: $d_{ij,t+1} = d_{ij,t} - 2\Delta t\, \hat{v}_{\max} \frac{\|\mathbf{F}'_{ij,t}\|_2}{\|\mathbf{F}'_{ij,t}\|_2 + \epsilon} < d_{ij,t}$.*

Proof. By Eq. (2), $\mathbf{F}'_{ij,t} = m_{ij}\, \mathbf{u}_{ij,t}$ with magnitude $m_{ij} = \exp\big(-\beta \frac{R_c - d_{ij,t}}{R_c - d_u}\big) > 0$ along $\mathbf{u}_{ij,t} = (\mathbf{p}_{j,t} - \mathbf{p}_{i,t})/d_{ij,t}$, while by anti-symmetry $\mathbf{F}'_{ji,t} = -m_{ij}\, \mathbf{u}_{ij,t}$. Under the isolation condition, $\mathbf{v}'_{i,t} = \tilde{v}\, \mathbf{u}_{ij,t}$ and $\mathbf{v}'_{j,t} = -\tilde{v}\, \mathbf{u}_{ij,t}$ where $\tilde{v} = \hat{v}_{\max} \frac{m_{ij}}{m_{ij} + \epsilon} \in (0, \hat{v}_{\max})$. Updating positions yields $\mathbf{p}_{i,t+1} - \mathbf{p}_{j,t+1} = (\mathbf{p}_{i,t} - \mathbf{p}_{j,t}) + 2\Delta t\, \tilde{v}\, \mathbf{u}_{ij,t} = -(d_{ij,t} - 2\Delta t\, \tilde{v})\mathbf{u}_{ij,t}$. Taking the Euclidean norm gives $d_{ij,t+1} = d_{ij,t} - 2\Delta t\, \tilde{v} < d_{ij,t}$. □

Remark 1 (Test-time safety margin under multi-neighbor interference and branch fusion)**.** *In dense swarms, competing neighbor forces and the learned velocity $\mathbf{v}''_{i,t}$ perturb this descent direction. Because the numerator of Eq. (4) is the convex combination $w_E(t)\mathbf{v}'_{i,t} + w_T(t)\mathbf{v}''_{i,t}$ of two vectors bounded in norm by $\hat{v}_{\max}$, preserving a non-negative projection along $\mathbf{v}'_{i,t}$ whenever $\mathbf{v}''_{i,t}$ opposes $\mathbf{v}'_{i,t}$ requires maintaining $w_E(t)$ above a state-dependent safety margin $w_E^{\min} \in (0, 0.5]$, or equivalently keeping $w_T(t)$ below $\tau = 1 - w_E^{\min} \in [0.5, 1)$. Under a steep schedule ($k = 30$), $w_E(t)$ decays exponentially toward zero after $t > s_0 H$, eliminating both local attractive capture and repulsive anti-clustering during the second half of long recovery trajectories.*

6.2 Fusion Schedule as an Implicit Training Curriculum

During end-to-end training, the sigmoid slope k governs how long the learned branch operates in an unassisted, high-autonomy regime. Beyond test-time link retention, the schedule $w_T(t; k, s_0)$ also shapes gradient propagation during the T_{train}-step rollout of Algorithm 1.

Definition 1 (High-autonomy exposure)**.** *For a horizon H and an autonomy threshold $\tau = 1 - w_E^{\min} \in (0.5, 1)$, the* high-autonomy exposure $\text{AE}(k, s_0; \tau)$ *of schedule $w_T(\cdot; k, s_0)$ is the fraction of steps in which the learned branch weight exceeds τ:*

$$\text{AE}(k, s_0; \tau) = \frac{1}{H} \Big| \Big\{ t \in \{0, \dots, H-1\} : w_T(t; k, s_0) > \tau \Big\} \Big|. \tag{6}$$

Inverting Eq. (3) shows that $w_T(t; k, s_0) > \tau$ if and only if

$$\frac{t}{H} > s_0 + \frac{1}{k} \ln\left(\frac{\tau}{1 - \tau}\right), \tag{7}$$

and the transition interval from $w_T = 0.1$ to $w_T = 0.9$ spans width $\Delta s_{10\to 90} = \frac{\ln(81)}{k} \approx \frac{4.394}{k}$ of the horizon. Equation (7) reveals the exact mathematical coupling between steepness k and midpoint s_0: for any safety threshold $\tau > 0.5$ (where $\ln \frac{\tau}{1-\tau} > 0$), reducing k directly delays the threshold crossing and widens $\Delta s_{10\to 90}$. Under R2C's steep default ($k = 30, s_0 = 0.125$), the transition window $\Delta s_{10\to 90}$ spans only 14.6% of the horizon and AE$(30, 0.125; 0.8)$ reaches 82.5%, forcing the GCN to control the swarm almost solo from $t/H = 0.17$ onward. Conversely, a smooth schedule ($k = 5, s_0 = 0.7$) expands $\Delta s_{10\to 90}$ to 87.9% of the horizon and reduces AE$(5, 0.7; 0.8)$ to 0.0%, keeping the learned branch in a well-conditioned residual-correction regime throughout both training and evaluation.

7 Experimental Setup

Simulation testbed and failure protocols follow R2C's physical parameters across random and uniform deployments. We evaluate all methods in a $1000 \times 1000\,\mathrm{m}^2$ mission square under two initial swarm layouts: **random deployment** with 200 UAVs sampled uniformly at random, and **uniform deployment** with 196 UAVs placed on a 14×14 regular grid. Following Peng et al. (2026), each UAV has communication and coverage radii $R_c = R_{\mathrm{cov}} = 120\,\mathrm{m}$, maximum speed $\hat{v}_{\max} = 1.0\,\mathrm{m/s}$, step size $\Delta t = 1.0\,\mathrm{s}$, and evaluation horizon $T = 400$ steps. At $t_1 = 0$, a failure ratio $r \in \{20\%, 40\%, 60\%, 80\%\}$ of UAVs is removed uniformly at random. Every architecture and ablation configuration is trained across 5 independent random seeds ($\texttt{seed} \in \{1, 2, 3, 4, 5\}$) and evaluated over 50 independent failure trials per seed (250 trials per configuration per failure rate).

Baselines and evaluation metrics quantify recovery reliability, speed, coverage, and clustering. We compare **HAC-Fixed** and **HAC-Smooth** ($k = 5, s_0 = 0.7$) against five baselines: (i) **GAC-Fixed** (global learned GCN), (ii) **SAC-Fixed** (subnetwork rigid-translation GCN), (iii) **Learned-only** ($w_T(t) \equiv 1.0$ ablation of HAC), (iv) **Heuristic-only** ($w_E(t) \equiv 1.0$ virtual-force branch), and (v) published reference numbers from **R2C** (Peng et al., 2026). We report four metrics: **Success/50** (number of trials out of 50 achieving $C_{t_r} = 1$ within $T = 400$ steps), **REC** (recovery time ratio T_{rec} averaged over successful trials), **COV** (coverage ratio $C_{\mathrm{cov}}(t_r)$ at recovery), and **AND** (trajectory-averaged node degree up to t_r, measuring spatial clustering).

Implementation details and strict multi-seed control ensure statistical reproducibility. All learned backbones use $L_{\mathrm{gcn}} = 4$ GCN layers, hidden width 128, ReLU activations, and a 2-layer MLP head, trained for 300 epochs using Adam (learning rate 10^{-3}, gradient norm clip 1.0) over $T_{\mathrm{train}} = 40$ rollout steps with 800 Monte Carlo coverage grid points per step. Both NumPy environment RNGs and PyTorch CPU/CUDA weight initializations are strictly seeded prior to model instantiation, and each seed $s \in \{1..5\}$ is evaluated on its own deterministic sequence of 50 test topologies.

8 Results

8.1 Main Results: HAC vs. Single-Branch Baselines

HAC-Fixed and HAC-Smooth achieve near-perfect recovery across 20%–60% failures and outperform single-branch learned baselines by up to 38.0/50 trials (76.0 pp) at 80% node loss. Tables 3 and 4 report the five-seed evaluation results under random and uniform deployments, and Figure 1 illustrates the success-rate distributions under random deployment. Across 20% to 60% failure rates in both geometries, HAC-Fixed achieves a 100% recovery rate (50.0 ± 0.0 out of 50 trials) with zero cross-seed variance, while HAC-Smooth achieves 49.6 to 50.0 recoveries

Table 3: Main results under random deployment (mean ± std over 5 seeds, 50 evaluation trials per seed). Success is out of 50 trials; n/a indicates zero successful trials.

Architecture	Failure	Success/50	REC	COV	AND
HAC-Fixed	20%	**50.0 ± 0.0**	0.051 ± 0.011	0.997 ± 0.000	6.73 ± 0.05
	40%	**50.0 ± 0.0**	0.134 ± 0.005	0.988 ± 0.001	5.35 ± 0.05
	60%	**50.0 ± 0.0**	0.331 ± 0.009	0.928 ± 0.008	4.01 ± 0.07
	80%	**46.6 ± 3.9**	0.696 ± 0.048	0.602 ± 0.009	2.58 ± 0.03
HAC-Smooth (5, 0.7)	20%	49.8 ± 0.4	**0.034 ± 0.008**	0.997 ± 0.000	6.63 ± 0.03
	40%	**50.0 ± 0.0**	**0.110 ± 0.007**	**0.993 ± 0.000**	5.22 ± 0.04
	60%	49.6 ± 0.5	**0.325 ± 0.020**	**0.949 ± 0.004**	3.86 ± 0.06
	80%	43.6 ± 2.8	**0.684 ± 0.020**	**0.616 ± 0.006**	**2.48 ± 0.06**
GAC-Fixed	20%	49.0 ± 0.6	0.324 ± 0.007	0.747 ± 0.009	13.68 ± 0.28
	40%	47.2 ± 1.7	0.467 ± 0.020	0.614 ± 0.021	12.48 ± 0.61
	60%	42.8 ± 3.7	0.612 ± 0.027	0.452 ± 0.022	10.55 ± 0.78
	80%	19.8 ± 5.5	0.775 ± 0.017	0.310 ± 0.009	6.00 ± 0.26
SAC-Fixed	20%	29.4 ± 3.4	0.232 ± 0.047	0.942 ± 0.014	6.71 ± 0.09
	40%	8.8 ± 1.9	0.577 ± 0.068	0.842 ± 0.037	5.32 ± 0.18
	60%	0.0 ± 0.0	n/a	n/a	n/a
	80%	0.0 ± 0.0	n/a	n/a	n/a
Learned-only	20%	45.6 ± 2.9	0.212 ± 0.025	0.901 ± 0.014	9.07 ± 0.44
	40%	37.8 ± 3.5	0.468 ± 0.030	0.719 ± 0.025	10.12 ± 0.33
	60%	26.4 ± 2.8	0.702 ± 0.020	0.503 ± 0.024	9.27 ± 0.46
	80%	8.6 ± 3.3	0.831 ± 0.023	0.347 ± 0.023	5.10 ± 0.33
Heuristic-only	20%	47.2 ± 1.2	0.045 ± 0.015	0.997 ± 0.000	6.64 ± 0.04
	40%	30.8 ± 1.3	0.125 ± 0.021	0.993 ± 0.001	5.14 ± 0.05
	60%	1.4 ± 1.4	0.227 ± 0.092	0.964 ± 0.007	3.78 ± 0.13
	80%	0.0 ± 0.0	n/a	n/a	n/a

with faster recovery times at 20%–60% random failure and higher post-recovery coverage (0.949 vs. 0.928 at 60% random failure; 0.967 vs. 0.960 at 60% uniform failure). At the extreme 80% failure rate under random deployment, HAC-Fixed succeeds in 46.6 ± 3.9 trials and HAC-Smooth succeeds in 43.6 ± 2.8 trials, surpassing GAC-Fixed (19.8 ± 5.5) by 26.8 trials (53.6 pp) and the Learned-only ablation (8.6 ± 3.3) by 38.0 trials (76.0 pp), while SAC-Fixed and Heuristic-only drop to 0.0 ± 0.0. Simultaneously, HAC-Smooth achieves the highest 80%-failure coverage (0.616 ± 0.006, vs. 0.602 ± 0.009 for HAC-Fixed and 0.310–0.347 for single-branch learned models) and lowest node clustering (AND $= 2.48 \pm 0.06$).

Uniform grid deployments confirm the same architectural ranking with even tighter variance. As shown in Table 4, both HAC-Fixed and HAC-Smooth achieve a 100% recovery rate (50.0 ± 0.0) across 20%, 40%, and 60% failures under uniform deployment. At 80% failure, HAC-Fixed reaches 49.4 ± 0.8 and HAC-Smooth reaches 47.8 ± 1.3 out of 50 trials, outperforming GAC-Fixed (25.8 ± 6.1) by up to 23.6 trials (47.2 pp) and Learned-only (11.2 ± 3.7) by 38.2 trials (76.4 pp) while nearly doubling their post-recovery coverage (0.604–0.609 vs. 0.300–0.344). Both SAC-Fixed and Heuristic-only collapse to 0.0 ± 0.0 recoveries at 80% failure across both deployment topologies.

8.2 Comparison with Published R2C Results

Without boundary hulls or multipartite graph convolutions, HAC retains 88.4% to 91.2% of R2C's reported coverage at 80% failure (within 5.9 to 7.9 pp). Table 5 compares five-seed means of HAC-Fixed and HAC-Smooth against R2C's published metrics (Peng et al., 2026). Because R2C's source code is unavailable, this comparison is indicative rather than a same-codebase reproduction. From 20% to 60% failure, HAC-Smooth matches R2C's coverage within 0.1 to 1.1 pp

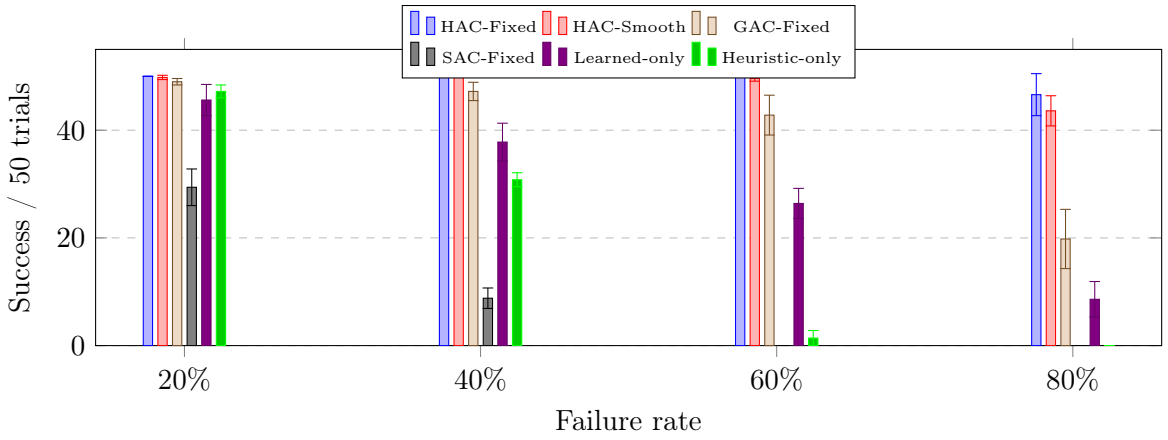


Figure 1: Connectivity recovery success count out of 50 trials across failure rates under random deployment (mean ± std over 5 seeds). Both HAC-Fixed and HAC-Smooth sustain high recovery rates across all failure regimes.

and achieves faster recovery at 20% and 40% uniform failure. At 80% failure, R2C's reported coverage exceeds HAC-Fixed by 7.9 pp under random deployment (0.681 vs. 0.602, or 88.4% relative retention) and by 5.9 pp under uniform deployment (0.668 vs. 0.609, or 91.2% relative retention), while HAC-Smooth narrows the random-deployment gap to 6.5 pp (0.681 vs. 0.616, or 90.5% relative retention). Considering that R2C employs Quickhull boundary extraction, cross-subgraph virtual edges, and an 8-layer, 512-width Chebyshev spectral GCN whereas HAC uses a single 4-layer, 128-width global k-NN GCN, these results demonstrate that velocity-level hybridization itself captures roughly 90% of the coverage benefit.

8.3 Topology Quality Analysis

HAC prevents GNN-induced spatial clustering while sustaining full-swarm reconnection. Figure 2 plots the five-seed trajectory-averaged node degree (AND) across failure rates under random deployment. Whereas GAC-Fixed and Learned-only produce heavily clustered topologies (9.07–13.68 neighbors), HAC-Fixed and HAC-Smooth track the low, smoothly decreasing degree profile of the non-clustering Heuristic-only and SAC-Fixed baselines (6.63–6.73 down to 2.48–2.58), confirming that velocity-level virtual-force repulsion counteracts GNN aggregation collapse throughout the recovery trajectory.

9 Ablation Studies

9.1 Branch Contribution Synergy

Velocity fusion yields super-additive recovery gains that exceed the sum of isolated branches at severe failure rates. Comparing HAC-Fixed against its isolated constituents in Tables 3 and 4 rules out simple branch selection. At 60% uniform failure, Heuristic-only recovers in 0.8 ± 0.7 trials and Learned-only in 36.4 ± 3.3 trials, whereas HAC-Fixed and HAC-Smooth recover in all 50.0 ± 0.0 trials. At 80% random failure, Heuristic-only (0.0 ± 0.0) and Learned-only (8.6 ± 3.3) sum to 8.6 recoveries, yet fusing them reaches 46.6 ± 3.9 (HAC-Fixed) and 43.6 ± 2.8 (HAC-Smooth). The global GCN pulls distant components into communication range R_c, where

Table 4: Main results under uniform grid deployment (mean ± std over 5 seeds, 50 evaluation trials per seed).

Architecture	Failure	Success/50	REC	COV	AND
HAC-Fixed	20%	$\mathbf{50.0 \pm 0.0}$	0.005 ± 0.001	1.000 ± 0.000	5.72 ± 0.01
	40%	$\mathbf{50.0 \pm 0.0}$	0.040 ± 0.004	0.998 ± 0.000	4.35 ± 0.02
	60%	$\mathbf{50.0 \pm 0.0}$	$\mathbf{0.249 \pm 0.023}$	0.960 ± 0.004	3.30 ± 0.04
	80%	$\mathbf{49.4 \pm 0.8}$	$\mathbf{0.643 \pm 0.026}$	$\mathbf{0.609 \pm 0.004}$	2.30 ± 0.03
HAC-Smooth $(5, 0.7)$	20%	$\mathbf{50.0 \pm 0.0}$	0.005 ± 0.001	1.000 ± 0.000	5.72 ± 0.01
	40%	$\mathbf{50.0 \pm 0.0}$	$\mathbf{0.039 \pm 0.004}$	0.998 ± 0.000	4.33 ± 0.02
	60%	$\mathbf{50.0 \pm 0.0}$	0.252 ± 0.019	$\mathbf{0.967 \pm 0.001}$	3.24 ± 0.02
	80%	47.8 ± 1.3	0.665 ± 0.040	0.604 ± 0.007	$\mathbf{2.28 \pm 0.03}$
GAC-Fixed	20%	50.0 ± 0.0	0.011 ± 0.003	0.992 ± 0.003	5.88 ± 0.04
	40%	50.0 ± 0.0	0.200 ± 0.014	0.843 ± 0.019	6.86 ± 0.18
	60%	47.0 ± 2.8	0.574 ± 0.013	0.478 ± 0.023	8.75 ± 0.42
	80%	25.8 ± 6.1	0.772 ± 0.057	0.300 ± 0.012	5.81 ± 0.31
SAC-Fixed	20%	48.8 ± 1.2	0.007 ± 0.007	0.999 ± 0.001	5.72 ± 0.01
	40%	31.4 ± 3.3	0.161 ± 0.013	0.959 ± 0.007	4.38 ± 0.04
	60%	0.2 ± 0.4	0.003 ± 0.000	0.964 ± 0.000	3.21 ± 0.00
	80%	0.0 ± 0.0	n/a	n/a	n/a
Learned-only	20%	49.6 ± 0.5	0.023 ± 0.015	0.988 ± 0.009	6.05 ± 0.30
	40%	48.4 ± 1.0	0.211 ± 0.014	0.892 ± 0.012	6.16 ± 0.25
	60%	36.4 ± 3.3	0.620 ± 0.068	0.569 ± 0.048	7.49 ± 0.92
	80%	11.2 ± 3.7	0.839 ± 0.039	0.344 ± 0.019	4.73 ± 0.55
Heuristic-only	20%	49.8 ± 0.4	0.006 ± 0.002	1.000 ± 0.000	5.72 ± 0.01
	40%	44.0 ± 1.3	0.078 ± 0.017	0.998 ± 0.000	4.36 ± 0.02
	60%	0.8 ± 0.7	0.110 ± 0.081	0.977 ± 0.009	2.87 ± 0.26
	80%	0.0 ± 0.0	n/a	n/a	n/a

the virtual-force branch's attractive zone (Proposition 1) locks links in place while its repulsive zone prevents over-concentration.

9.2 Adaptive Schedule Steepness and Train–Evaluation Decoupling

Across five seeds, reducing sigmoid steepness k monotonically improves recovery, coverage, and node clustering at every transition midpoint s_0. Table 6 reports the five-seed schedule steepness sweep alongside theoretical transition width $\Delta s_{10\to 90} = \frac{\ln 81}{k}$ and high-autonomy exposure $\mathrm{AE}(k, s_0; 0.8)$ computed via Eq. (6). Unlike single-seed preliminary runs, five-seed averaging reveals a strictly monotonic relationship between transition smoothness and recovery performance at fixed s_0: at $s_0 = 0.125$, reducing k from 30 to 10 improves 60%-failure recovery from 41.0 ± 3.3 to 46.0 ± 2.2 and 80%-failure recovery from 14.2 ± 4.3 to 18.4 ± 5.0 while lowering AND from 4.11 to 3.43. At $s_0 = 0.5$, reducing k from 30 to 10 boosts 80%-failure recovery nearly six-fold from 5.2 ± 1.3 to 29.6 ± 4.5. Finally, **HAC-Smooth** ($k = 5, s_0 = 0.7$) expands $\Delta s_{10\to 90}$ to 87.9% and reduces $\mathrm{AE}(5, 0.7; 0.8)$ to 0.0%, reaching 43.6 ± 2.8 recoveries at 80% random failure (47.8 ± 1.3 under uniform deployment) with the highest coverage (0.616 ± 0.006) and lowest clustering (2.48 ± 0.06).

Decoupling training and evaluation schedules proves that test-time heuristic safety margins prevent rollout collapse while smooth training curricula minimize variance. To separate the test-time safety-margin effect (Remark 1) from the training-time curriculum effect (§6.2), Table 7 evaluates checkpoints trained under steep $(30, 0.125)$, fixed (0.5), and smooth $(5, 0.7)$ schedules when deployed under fixed equal weights ($w_E^{\text{eval}} = w_T^{\text{eval}} = 0.5$). Remarkably, taking the exact checkpoint trained under R2C's steep schedule $(30, 0.125)$ and evaluating it with fixed 50/50 weights jumps its 80%-failure recovery count from 14.2 ± 4.3 to 46.6 ± 0.5 while raising

Table 5: Comparison between HAC (5-seed means) and R2C's published numbers (Peng et al., 2026). At 80% failure, HAC comes within 6.5–7.9 pp (random) and 5.9–6.4 pp (uniform) of R2C's coverage while omitting subnetwork boundary extraction and multipartite graph construction.

	Random Deployment (REC / COV)			Uniform Deployment (REC / COV)		
Failure	R2C	HAC-Fixed	HAC-Smooth	R2C	HAC-Fixed	HAC-Smooth
20%	0.023 / 0.996	0.051 / 0.997	0.034 / 0.997	0.037 / 1.000	0.005 / 1.000	0.005 / 1.000
40%	0.070 / 0.989	0.134 / 0.988	0.110 / 0.993	0.051 / 0.999	0.040 / 0.998	0.039 / 0.998
60%	0.204 / 0.951	0.331 / 0.928	0.325 / 0.949	0.181 / 0.978	0.249 / 0.960	0.252 / 0.967
80%	0.625 / **0.681**	0.696 / 0.602	0.684 / **0.616**	0.617 / **0.668**	0.643 / **0.609**	0.665 / 0.604

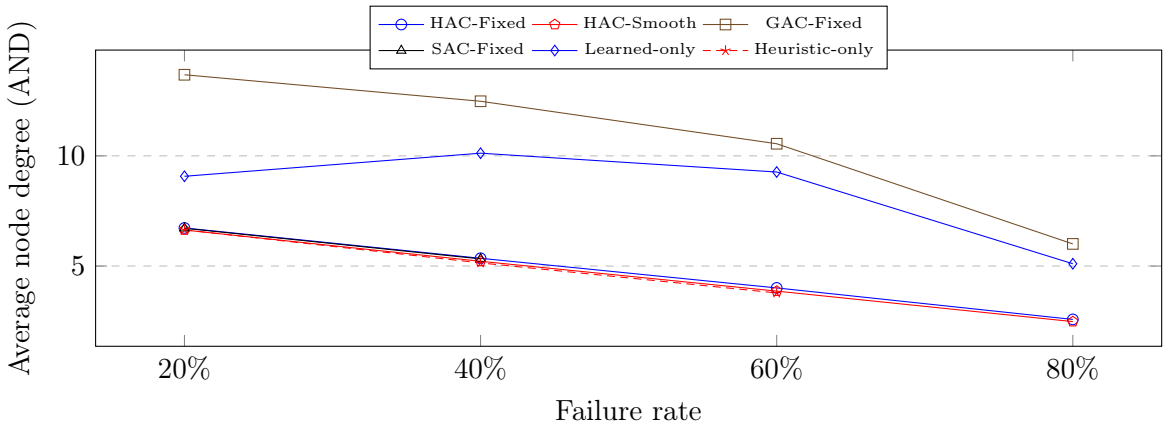


Figure 2: Trajectory-averaged node degree (AND) vs. failure rate under random deployment (5-seed means). HAC-Fixed and HAC-Smooth match the low clustering profile of SAC-Fixed and Heuristic-only while avoiding their high-failure connectivity collapse.

coverage from 0.413 to 0.622 ± 0.004 and cutting AND from 4.11 to 2.51. This confirms that the learned GCN branch acquires a valid global attraction field even under steep training, and that the primary collapse under $(30, 0.125)$ occurs at *evaluation time* when $w_E(t) \to 0$ removes heuristic link capture and local repulsion. Moreover, pairing smooth curriculum training $(5, 0.7)$ with fixed 50/50 evaluation (**HAC-Curriculum**) achieves the highest overall 80%-failure recovery rate (47.4 ± 1.9 out of 50, with 50.0 ± 0.0 across 20%–60% failure) and cuts cross-seed variance in half compared to fixed training (1.9 vs. 3.9).

9.3 Granularity Substitution: HSC as a Counter-Experiment

Schedule smoothing rescues subnetwork-level hybrids at moderate fragmentation (60% failure) but cannot overcome rigid-translation expressiveness limits at 80% failure. We evaluate **HSC** across five seeds by fusing the virtual-force heuristic branch with SAC's subnetwork-level rigid-translation GCN. As shown in Table 8, at 60% failure, replacing the steep schedule $(30, 0.125)$ with the smooth schedule $(5, 0.7)$ rescues HSC from 12.2 ± 3.7 to 40.8 ± 1.5 recoveries under random deployment (surpassing HSC-Fixed's 38.0 ± 2.7) and from 13.0 ± 4.9 to 45.0 ± 2.8 under uniform deployment. However, at 80% failure, all three HSC variants collapse to 0.0 ± 0.0–0.4 ± 0.5 recoveries across both deployments. This five-seed counter-experiment cleanly demarcates the boundary between scheduling and expressiveness: when multi-node subnetworks remain intact at 60% failure, smooth scheduling restores hybrid coordination, but when 80% node loss shatters the swarm into isolated singletons and tiny clusters, rigid subnetwork translation lacks the node-level

Table 6: Five-seed schedule steepness sweep under random deployment (mean ± std over 5 seeds), showing transition width $\Delta s_{10\to 90}$, high-autonomy exposure AE$(k, s_0; 0.8)$, recovery counts (Success/50), and 80%-failure COV and AND.

Schedule (k, s_0)	$\Delta s_{10\to 90}$	AE(0.8)	40% Succ	60% Succ	80% Succ	80% COV	80% AND
R2C Default $(30, 0.125)$	14.6%	82.5%	47.2 ± 1.6	41.0 ± 3.3	14.2 ± 4.3	0.413 ± 0.032	4.11 ± 0.30
Intermediate $(10, 0.125)$	43.9%	72.5%	49.4 ± 0.8	46.0 ± 2.2	18.4 ± 5.0	0.462 ± 0.036	3.43 ± 0.32
Delayed Steep $(30, 0.5)$	14.6%	45.0%	49.6 ± 0.5	42.8 ± 0.7	5.2 ± 1.3	0.516 ± 0.020	2.60 ± 0.15
Delayed Smooth $(10, 0.5)$	43.9%	35.0%	50.0 ± 0.0	49.8 ± 0.4	29.6 ± 4.5	0.571 ± 0.022	2.58 ± 0.08
HAC-Smooth $(5, 0.7)$	**87.9%**	**0.0%**	$\mathbf{50.0 \pm 0.0}$	$\mathbf{49.6 \pm 0.5}$	$\mathbf{43.6 \pm 2.8}$	$\mathbf{0.616 \pm 0.006}$	$\mathbf{2.48 \pm 0.06}$

Table 7: Train–evaluation schedule decoupling study at 80% failure under random deployment (mean ± std over 5 seeds). Restoring $w_E = 0.5$ at evaluation time rescues the steep-trained policy (+32.4 trials), while smooth curriculum training combined with fixed evaluation (**HAC-Curriculum**) achieves the highest recovery rate (47.4 ± 1.9) and faster reconnection.

Training Schedule w_T^{train}	Evaluation Schedule w_T^{eval}	80% Success/50	80% REC	80% COV	80% AND
Steep $(30, 0.125)$	Steep $(30, 0.125)$	14.2 ± 4.3	0.796 ± 0.032	0.413 ± 0.032	4.11 ± 0.30
Steep $(30, 0.125)$	Fixed (0.5)	46.6 ± 0.5	0.679 ± 0.023	$\mathbf{0.622 \pm 0.004}$	2.51 ± 0.05
Fixed (0.5) [HAC-Fixed]	Fixed (0.5)	46.6 ± 3.9	0.696 ± 0.048	0.602 ± 0.009	2.58 ± 0.03
Smooth $(5, 0.7)$ [HAC-Smooth]	Smooth $(5, 0.7)$	43.6 ± 2.8	0.684 ± 0.020	0.616 ± 0.006	$\mathbf{2.48 \pm 0.06}$
Smooth $(5, 0.7)$ **[HAC-Curriculum]**	**Fixed** (0.5)	$\mathbf{47.4 \pm 1.9}$	$\mathbf{0.674 \pm 0.027}$	0.602 ± 0.005	2.58 ± 0.03

degrees of freedom required to reconnect distant components.

10 Discussion

10.1 Underlying Physical and Optimization Mechanisms

Complementary receptive fields, test-time heuristic safety margins, and residual curriculum supervision explain the effectiveness of simple global hybrids. Across our five-seed studies, the heuristic virtual-force branch and the global k-NN GCN address orthogonal spatial scales: the global GCN supplies long-range attraction gradients across disconnected components separated beyond R_c, while the virtual-force branch enforces local collision buffering ($d < d_l$) and certifiable link capture ($d_u < d \leq R_c$, Proposition 1). Our train–evaluation decoupling study (Table 7) demonstrates that maintaining $w_E(t) \geq w_E^{\min}$ throughout deployment is essential to prevent the learned branch from breaking newly formed links, while training under a smooth schedule ($k = 5$) stabilizes gradient rollouts and halves cross-seed variance.

10.2 Practical Deployment Guidelines

Practitioners should match fusion schedules to policy expressiveness and expected failure severity. Our findings translate into three actionable design rules for resilient multi-UAV controllers:

- **Combine smooth curriculum training with fixed 50/50 evaluation (HAC-Curriculum) for peak high-damage reliability:** Training with $(k = 5, s_0 = 0.7)$ and deploying with $w_E = w_T = 0.5$ achieves 100% recovery across 20%–60% failures and the highest 80%-failure recovery rate ($47.4 \pm 1.9/50$) with low variance.
- **Use end-to-end HAC-Smooth** $(k = 5, s_0 = 0.7)$ **when post-recovery coverage uniformity is paramount:** Deploying the smooth schedule directly achieves faster recovery at

Table 8: Five-seed counter-experiment with HSC (heuristic branch + subnetwork-level rigid-translation GCN, mean ± std over 5 seeds). Schedule smoothing rescues 60%-failure recovery but cannot overcome HSC's lack of node-level expressiveness at 80% failure.

	Random Deployment (Success/50)			Uniform Deployment (Success/50)		
Variant	40%	60%	80%	40%	60%	80%
HSC-Fixed	47.6 ± 1.5	38.0 ± 2.7	0.4 ± 0.5	49.6 ± 0.5	$\mathbf{46.4 \pm 1.4}$	0.2 ± 0.4
HSC-Adaptive $(30, 0.125)$	47.2 ± 1.5	12.2 ± 3.7	0.0 ± 0.0	49.4 ± 0.5	13.0 ± 4.9	0.0 ± 0.0
HSC-Smooth $(5, 0.7)$	$\mathbf{49.2 \pm 0.7}$	$\mathbf{40.8 \pm 1.5}$	0.0 ± 0.0	$\mathbf{49.8 \pm 0.4}$	45.0 ± 2.8	0.0 ± 0.0

light-to-moderate damage and improves 60%–80% coverage (0.949 and 0.616 vs. 0.928 and 0.602 for HAC-Fixed) with lower node clustering (AND 2.48 vs. 2.58).

- **Ensure node-level output expressiveness before expecting resilience above 60% node loss:** Subnetwork-level rigid translation (SAC, HSC) can be stabilized by smooth scheduling up to 60% failure (40.8–45.0/50), but requires per-node velocity residuals to survive 80% fragmentation.

10.3 Limitations and Future Work

Closing the remaining 5.9 to 7.9 pp high-failure coverage gap and scaling training horizons remain open directions. First, while HAC retains roughly 90% of R2C's reported coverage at 80% failure, isolating whether R2C's remaining 5.9–7.9 pp advantage stems from Quickhull boundary weighting, multipartite virtual edges, or its 16× wider spectral GCN requires a controlled re-implementation of those sub-modules inside our simulator. Second, all learned policies are trained on $T_{\text{train}} = 40$ rollout steps for backpropagation-through-time tractability and evaluated over $T = 400$ steps; curriculum horizon scaling or truncated BPTT over longer rollouts could further improve late-stage coverage.

11 Conclusion

We systematically deconstructed the hybrid design space for UAV swarm connectivity recovery along decision paradigm, spatial reasoning granularity, and trade-off scheduling across five random seeds. By pairing a parameter-free virtual-force branch with a single global k-NN graph convolution branch, HAC achieves near-perfect connectivity restoration across 20% to 80% node failures, outperforms single-branch learned baselines by up to 76.4 pp in recovery rate, and retains 88.4% to 91.2% of R2C's reported coverage without subnetwork partitioning or convex-hull boundary extraction. Through five-seed schedule steepness sweeps, train–evaluation schedule decoupling, and a subnetwork-level counter-experiment (HSC), we proved that smooth adaptive schedules provide a dual benefit—preserving local heuristic safety margins at test time and regularizing end-to-end rollout training—provided the learned policy possesses node-level velocity expressiveness.

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A Deviations and Bug-Fix Log

We document every implementation choice where our testbed deviates from Peng et al. (2026) or corrects numerical issues discovered during development:

- **Failure-node selection.** R2C specifies "unpredictable random failure patterns" without detailing the sampling distribution; we apply uniform random node removal without replacement.
- **Count-invariant connectivity surrogate loss.** R2C's discrete component count objective is non-differentiable. An initial soft-minimum estimator omitted the $-\log(n_i n_j)$ normalization in Eq. (5), causing large component pairs to produce spuriously negative losses purely due to pair-count summation bias. On a synthetic test with true minimum separation ≈ 100–$108\,\mathrm{m}$,

the uncorrected estimator returned 2.85 for 3×3 nodes vs. -230.58 for 30×30 nodes, whereas our corrected Eq. (5) returned 112.71 and 109.54, respectively. All reported models use the corrected loss.

- **Boundary buffer width and clamping.** R2C leaves d_e unspecified; we set $d_e = 0.15R_c = 18\,\text{m}$ and enforce $0 \leq d_{i,b} < d_e$ so nodes clamped exactly onto mission boundaries continue to receive inward repulsion.
- **Rollout horizon and Monte Carlo coverage.** Differentiable training uses $T_{\text{train}} = 40$ steps with 800 uniform random grid points per step, whereas evaluation runs for $T = 400$ steps on a deterministic 5 m grid (40,000 points).
- **Full-stack training and evaluation seeding.** Both NumPy environment RNGs and PyTorch CPU/CUDA weight initializations are seeded via `-seed` in both training and evaluation scripts. Earlier unseeded GAC-Fixed runs yielded an optimistic 20%-failure COV of 0.830 and AND of 11.87, whereas five properly seeded runs yield COV 0.747 ± 0.009 and AND 13.68 ± 0.28 (Table 3). All tables in this paper report five-seed means and standard deviations under strict full-stack seeding.

B Hyperparameter Summary

Table 9: Hyperparameters shared across all trained architectures unless otherwise noted.

Parameter	Value
GCN layers L_{gcn} / Hidden dimension	4 / 128
Euclidean k-NN degree k_{nn}	8
Optimizer / Learning rate / Gradient clip	Adam / 1×10^{-3} / 1.0
Training epochs / Training rollout T_{train}	300 / 40 steps
Evaluation horizon T / Step size Δt	400 steps / 1.0 s
Movement cost coefficient λ	0.01
Soft-min temperatures $\tau_{\text{c}}, \tau_{\text{v}}$	50.0 m, 10.0 m
Heuristic decay constants α, β, δ	0.01
Neutral zone $[d_l, d_u]$ / Boundary buffer d_e	$[0.8R_c, 0.9R_c]$ / $0.15R_c$
Communication & coverage radii $R_c = R_{\text{cov}}$	120 m
Maximum UAV speed $\hat{v}_{\max}$	1.0 m/s
Mission area $\mathcal{A}$	$1000 \times 1000\,\text{m}^2$
Swarm size (Random / Uniform 14×14)	200 / 196 UAVs
Independent random seeds $\times$ Eval trials per seed	$5 \times 50 = 250$ trials

C Code Structure and Full Reproduction Suite

The accompanying repository provides end-to-end training, evaluation, and five-seed aggregation scripts for all architectures and ablations reported in Tables 2–8, together with the aggregated five-seed summary CSVs in `code/results/summary_multiseed_all.csv`.